

AASTHA PATEL

Computer Science Engineer

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ABOUT ME

Computer Science graduate with a strong foundation in Artificial Intelligence, Machine Learning, and Data Science. Passionate about building intelligent applications and solving real-world problems through AI-driven solutions. Currently expanding my expertise in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, and Agentic AI while continuously strengthening my skills in modern AI development.

TECHNICAL SKILLS

Programming Language: Python, SQL, C/C++, Java , PostgreSQL.

Framework & libraries: TensorFlow, Keras, OpenCV, Scikit-Learn, Pandas, NumPy, Flask, FastAPI, Pydantic, Matplotlib, Seaborn , SQLAlchemy , Axios , React

Tools & Platform: Linux, Git & Github, MLflow, Weights & Biases, MySQL, MongoDB, Power BI, VS Code, Google Colab, Jupyter Notebook

EDUCATION

Degree	Specialization	Institute	Year
B.Tech	Computer Science	Indus University	2022 - 2026
HSC	-	Shree Gayatri Vidhayala	2021 - 2022
SSC	-	Seventh day Adventist Higher Secondary School	2019 - 2020

PROJECT

AGROSENSE - AI

January , 2026 – May , 2026

- Developed an AI-powered crop disease detection web application using **Python, Flask, TensorFlow, and OpenCV**.
- Implemented CNN-based deep learning models to classify crop diseases from leaf images with high accuracy.
- Provided treatment and prevention recommendations based on the predicted disease.
- Integrated a simple and responsive web interface for real-time disease prediction and user interaction.

AI-POWERED-BRAIN-TUMOR-DETECTION-SYSTEM

July 2025 – December - 2025

- Developed an AI-based MRI Brain Tumor Detection System using a trained CNN model (Keras/TensorFlow).
- Built a full-stack Flask web application with secure MRI upload, preprocessing, and real-time tumor prediction.
- Designed a responsive frontend using HTML, CSS, JavaScript, featuring drag-and-drop uploads and animated results.
- Integrated PDF report generation, MRI preview, and confidence scoring for analysis interpretation.
- Utilized Git LFS to manage large model files and optimize deployment.
- Conducted model training, evaluation, and visualization through Jupyter Notebooks using medical imaging datasets.

SMARTPASS - AI ENTERPRISE VISITOR MANAGEMENT SYSTEM

JUNE 2026

- Developed a full-stack visitor management system using React, FastAPI, and PostgreSQL.
- Implemented JWT authentication, role-based access, and secure REST APIs.
- Built features for visitor registration, search, QR code generation, check-in/check-out, and history management.
- Designed a responsive admin dashboard with a reusable component architecture.
- Planned AI enhancements including OCR-based ID scanning, face verification, and visitor analytics.

CERTIFICATIONS

- Introduction And Evolution of Futuristic Technologies with applications , Indus University
- TensorFlow for Deep Learning Bootcamp , Udemy
- Complete Generative AI course with LangChain and HuggingFace , Udemy
- Complete AI & Machine Learning Data Science Bootcamp , Udemy
- Machine Learning with python , freecodecamp
- Complete Core Java in Simple Way , Udemy
- Innovate With Full Stack , Indus University